Model plan comments

Agile research

It is very recommended to do the project using Agile research methodology.
Do a small step, get feedback and continue.
Instead of jumping in depth in a single model, build a simple version of all of them, evaluate the performance and use it to decide where to invest more time.

Some of you plan to enhance the data, which is great and encouraged. Using agile research, you can estimate if the effort in the data enhancement is likely to be beneficial. Remember that some data, like production company, might not be available for many movies, reducing its value.

Most of you planned simple models, around a single table, with few parameters. These are great models to start with, fast to implement and easy to understand their errors (see error analysis in the project description)

Lighthouse metric

Some metrics can be taken into extreme (e.g., getting 100% recall by always recommending). If you optimize such a metric, add a constraint (e.g., minimal precision of x%, and justify the value of x).
Use metric that you will be able to easily implement in the database (e.g., precison@20, might require investment).
You already have an implementation of the confusion matrix.

Metrics in models

You can use any metric to improve your model.
You don't have to restrict yourselves to counts and Jaccard demonstrated in class.
The team that plans to use Laplace - well done!

In order to know model performance we should evaluate it. However, developing intuition can save you a lot of time.

Each movie has probably only a few good recommendations. If your model matches many movies to it (e.g., close years, same genre) you will have very low precision.

When matching movies by actors, an actor that played in only two movies might indicate a strong match or accidental errors.
An actor that played in many movies might be a star, indicating a strong connection or a matcher of unrelated movies.
When we have opposing intuition, quick evaluations do help.

Most movies do not have too many genres. Using Jaccard on the number of matching genres should return the same results in most cases, making it less beneficial.

The secret to high performance is usually in the data manipulation. Try to understand where your model makes unrealistic assumptions (e.g,, all roles are of similar importance, better to ignore “himself” roles instead of using “x as himself”, treating policeman and police officer as different roles). Handling these cases is usually very beneficial.

Collaborative filtering

Collaborative filtering (“People that like X, also liked Y) allows you to jump above the strict paths of directors, actors, etc. It is very recommended to have at least one such model.

Aggregation

While you can aggregate the vote of your model simply by using count/sum/avg, it is wasting performance. The textbook recommends learning the weight per model (e.g., using linear regression) yet this is not required. However, estimating weight per model is helpful.

Also, you do not have to restrict yourselves to binary votes. Your model can provide confidence levels (e.g., sharing 6 actors is a stronger match than sharing 3 actors).